

Student Information

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Overall Result

24/28 (85.7%) Grade: B

Score by Section

Section	Score	Pct
Activation Functions	5/5	100.0%
Weight Initialization	3/3	100.0%
Loss Functions	2/3	66.7%
Optimizers	3/4	75.0%
Backpropagation	2/3	66.7%
Regularization	3/4	75.0%
Batch Normalization	2/2	100.0%
Architecture Design	4/4	100.0%

Questions to Review

Q09. [Loss Functions]

Which loss function should you use for BINARY classification?
Binary Cross-Entropy = $-[y \cdot \log(p) + (1-y) \cdot \log(1-p)]$. Designed for 2-class problems with a Sigmoid output.

Q13. [Optimizers]

Adam optimizer combines which two techniques?
Adam maintains m (first moment, like momentum) and v (second moment, like RMSProp). Update: $\theta \leftarrow \theta - \eta \cdot \hat{m} / (\sqrt{\hat{v}} + \epsilon)$.

Q18. [Backpropagation]

In a gradient flow chart, what does a gradient ratio greater than 1000x between layers indicate?
1000x difference means early layers receive almost no learning signal. Fix: use ReLU, BatchNorm, or residual connections.

Q21. [Regularization]

L2 regularization adds λ multiplied by the sum of squared weights to the loss. What effect does this have?

*L2 adds λW to each gradient update, decaying weights toward 0 but never exactly 0.
Also called weight decay.*

